

Organization and Management Exam Notes

ME 708 · IV-I (BEL, BEX, BCT, BEI) · TU, IOE · tiers over 29 papers, 2065–2082 BS

4. Case Studies

2 hrs · asked in **14 of 29 papers** · 5 syllabus topics · **92 marks, 3.9% of the archive** → small, but a near-certain 4-mark short note since 2079.

READ THIS FIRST

Only two things are ever asked, in rotation: **objectives of case study** (8 papers) and **steps / phases of case study** (5 papers, all since 2079). Learn the **6 steps** and the **5 phases** as two separate numbered lists; examiners use both words. Twice a paper has set an applied 8-marker (prepare a case study of NEA or Nepal Telecom; recommend improvements for your college): answer it by filling in the **report structure** of §4.3. §4.4 and §4.5 are those two cases **written out in full**, as models. §4.6 is a **complete sample report** of an industrial visit: the 10-mark internal assessment report.

Tier chips count **papers out of 29**: **TOP n/29** 8+ **HOT n/29** 4–7 **PIN n/29** 1–3 inside an 8-mark question. Bold year = Regular exam.

4.1 Introduction and Objectives

What is a case study? Why are case studies conducted? Objectives and value of case study **2 3 4**

TOP 8/29 81 Ba, 80 Ba, 73 Ch, 71 Ch, 71 Shr, 70 Asa, 69 Ch, 66 Bh · short note in 71 Shr, 69 Ch, 66 Bh ("value of case study")

Meaning

- A **case** = written description of a **real management situation**: facts, reports, even opinions, but **no evaluation**.
- **Case study method**: learners analyse the case, identify problems, weigh alternatives and **decide**, as a manager would.
- Belief behind it: managerial competence comes from studying and discussing **actual situations**, not only theory.
- Word means different things to different users:
 - lawyer: a legal case;
 - researcher: a vehicle for **testing a hypothesis**;
 - management education: an **authentic problem** to train analysis and decision making.
- Aim secured by facing the learner with a **whole series** of cases, not one.
- Harvard Business School made it the standard method of management teaching (1920s) **verified/added**.

Why conducted (AJ Sir)

- Teach managerial skills: **analyse** a situation, **frame** the right questions, **find answers**, **make decisions**.
- Link the three foundations of management practice: **research, theory, practice**.

Objectives

- 1. Impart knowledge and facts** about real organizations.
- 2. Develop problem analysis skill**: separate symptoms from causes.
- 3. Develop decision-making skill** under incomplete information.
- 4. Improve communication**: discuss, defend, write reports.
- 5. Shape attitude**: nothing in human behaviour is absolutely right or wrong; respect other views.
- 6. Apply theory to practice**: test principles against reality.
- 7. Build teamwork** and judgement through group discussion.

Value (short note, 66 Bh)

- + realistic, active learning.
- + develops analytical thinking; learner participates.
- + many solutions welcome → no single “right” answer.
- + bridges classroom and industry.
- – time consuming: gathering, arranging and writing up the facts is long and tedious.
- – needs a skilled discussion leader.
- – one case may not generalize.

Types of case studies 2

PIN 1/29 80 Ba

By research purpose

- **Illustrative:** descriptive; one or two instances show what a situation is like; makes the unfamiliar familiar; gives readers a common language.
- **Exploratory (pilot):** short study before a large investigation, to find the right questions and measures. Risk: early findings released as conclusions.
- **Cumulative:** combines past studies from several sites and times → wider generalization at no new cost.
- **Critical instance:** one or a few sites of unique interest, or to challenge a universal claim; answers cause-effect questions.

4.2 Phases and Steps of Case Study

Explain the phases and steps of case study, with an example from your field 4 5 8

HOT 5/29 82 Bh, 81 Bh, 81 Ba, 80 Bh, 79 Bh · short note in 82 Bh, 81 Bh · every paper since 2079 except 80 Ba

Five phases (group discussion)

Five phases (group discussion)



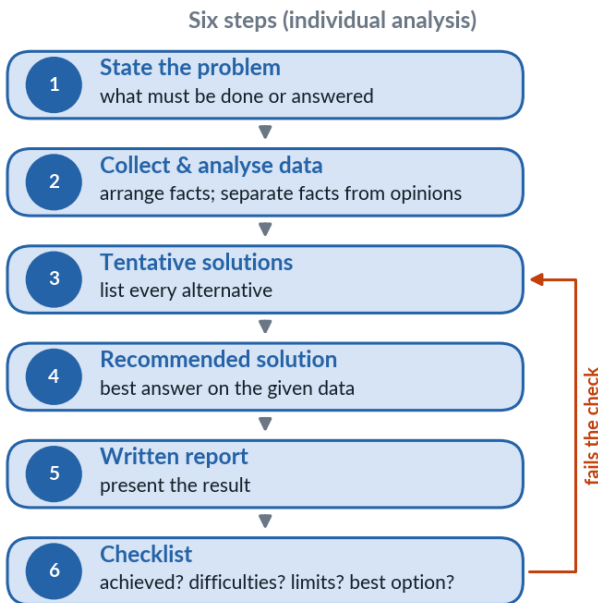
Case study vs lecture verified/added

	Lecture	Case study
Learner	listens	analyses, decides
Content	principles	real situation
Answer	one correct	several defensible
Skill built	recall	judgement, communication

Teaching cases

- **Incident:** a short event to react to.
- **Problem / decision:** choose an action.
- **Appraisal:** evaluate a situation.
- **Descriptive:** shows how a firm handled something.

Six steps (individual analysis)



Notes on the steps

- **Step 1:** read the case and its questions, then write in one line what the case is about and what action or answer is required.
- **Step 2:** arrange facts systematically; mark which are facts and which are opinions of people in the case.
- **Step 3:** treat first ideas as tentative; give every alternative due consideration before rejecting it.
- **Step 4:** choose **one** solution, the best answer on the given data.
- **Step 6 checklist:** what does it achieve? what difficulties? what limits (time, money, equipment, manpower)? is it the best alternative?
- A solution that fails the checklist sends you back to step 3.

Phases vs steps: which to write

- **Phases** = how a **group** works through a case in a class or workshop (discussion method).
- **Steps** = how **one analyst** solves it and writes it up (report method).
- Question says “phases” → 5 phases; “steps” → 6 steps; “phases and steps” (81 Bh) → both lists, 2 marks each.
- Always add one line of example from engineering (below).

Analysing a case problem (AJ Sir): the reading order before step 1

1. Read the **title** of the case.
2. Note the **company** and the case **problems** as given.
3. **Careful, detailed reading** of the whole case.
4. **Meet the central characters:** who decides, who is affected.
5. **Define the problems**, then develop the solution.

Example from engineering: why did a hydropower project finish two years late? verified/added

Step	Applied
1 State the problem	60 MW project, planned 4 years, took 6; cost overrun 40%. Why, and how to prevent it next time?
2 Collect and analyse data	Schedule vs actual (Gantt), change orders, land-acquisition dates, contractor payments, site logs, interviews with engineers and locals. Finding: land dispute (8 months), tunnel geology surprise (10 months), late payments → contractor slowdown.
3 Tentative solutions	(a) better geological survey; (b) settle land before tender; (c) escrow payment to contractor; (d) matrix project organization with one accountable manager.
4 Recommended solution	Combine (a)+(b) as pre-construction gates, (c) for cash flow; (d) as the structure.
5 Written report	Background, problem, data, analysis, recommendations, implementation plan.
6 Checklist	Achieves on-time completion; needs upfront survey budget; limit: political land issues; still the best alternative.

4.3 Case Report Writing, and Preparing a Case Study

Prepare a case study of an organization (NEA / Nepal Telecom; your college) following the case structure 8

PIN 2/29 74 Ch, 72 Ch · applied 8-markers: fill in the structure · full models in §4.4 (72 Ch) and §4.5 (74 Ch)

Case report writing methodology

- 1. Organizational background:**
 - goals and objectives; formation;
 - structure; social needs and value;
 - historical relation; financial position.
- 2. Identification of the problem.**
- 3. Hypothesis:** the suspected cause.
- 4. Case research design:** what data, from whom, how.
- 5. Collection and analysis of data.**
- 6. Generalization and interpretation.**
- 7. Conclusions and recommendations.**

Typical report titles (a full O&M case study)

- 1.** Acknowledgement.
- 2.** Organization: history, objectives, input, process, output.
- 3.** Organization structure (chart).
- 4.** Form of ownership.
- 5.** Personnel policies; manpower planning.
- 6.** Recruitment and selection; training and development.
- 7.** Job evaluation; merit rating; wages and incentives.
- 8.** Motivation; marketing; leadership; MIS.
- 9.** Drawbacks; suggestions and recommendations.
- 10.** Conclusion; references.
 - The titles are the syllabus itself: a case study is where Ch. 1–3 and 5 are applied to one real organization.

Writing the 8-mark version in the exam

- Use the **7 headings** of the methodology as your headings; 1–3 lines each.
- Spend the marks on **what the question names**: 72 Ch names planning horizon, leadership, motivation, HRD → one analysis paragraph each.
- Give **one or two real numbers** (loss %, load-shedding hours, pass rate); they show the data step was done.
- Draw **one** figure: an org chart, a cause-and-effect (fishbone) diagram, or a time-bound plan.
- End with **recommendations + time frame**, then the **checklist** line (achieves, difficulties, limits, best alternative).
- Roughly: background 1, problem + hypothesis 1, design + data 2, analysis 2, recommendations 2 marks.

4.4 Mock Case Study 1: Nepal Electricity Authority (72 Ch)

PIN 1/29 72 Ch · written out in full as a model · figures are NEA's published figures; staff-survey results are illustrative verified/added

Question (72 Ch, 8 marks)

If you are asked to prepare the case study considering the planning horizon, leadership, motivation and human resource development for either Nepal Electricity Authority or Nepal Telecom to improve the existing performance of these institutions, how do you prepare the case study following its structure?

Title

Improving the performance of Nepal Electricity Authority: a case study of planning horizon, leadership, motivation and human resource development. Period studied: FY 2072/73 (the crisis year) to FY 2080/81.

Acknowledgement

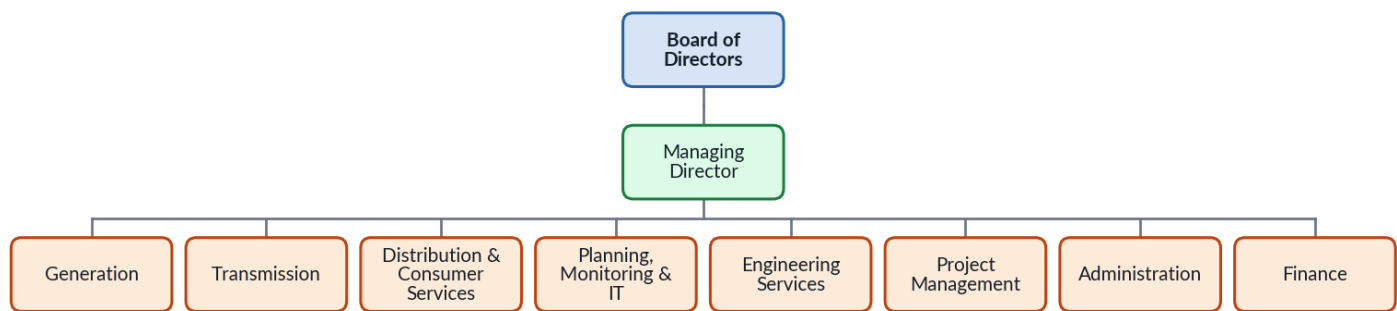
Thanks to NEA's Planning, Monitoring and IT Directorate for the annual reports, to the distribution-centre chiefs who gave interviews, and to the course teacher for guidance.

1. Organizational background

- **Formation:** established 16 August 1985 (Bhadra 1, 2042) under the **NEA Act 2041**, merging the Electricity Department, Nepal Electricity Corporation and related development boards.
- **Form of ownership:** **public corporation**, fully government owned (§1.4): own legal entity, govt control, service + commercial motive.
- **Goals:** generate, transmit and distribute adequate, reliable and affordable power; plan and develop the national grid; trade power with India.
- **Structure:** Board of Directors chaired by the Energy Minister → Managing Director → Deputy Managing Directors heading functional directorates (figure) → regional offices and distribution centres. A **line and staff** structure (§1.5).
- **Social value:** power drives industry, health, education, irrigation, telecom; every household is a customer.
- **Financial position:** accumulated loss of **Rs 34.61 bn** up to FY 2072/73; first net profit **Rs 1.46 bn** in FY 2073/74, profitable every year since.

Input, process, output

Stage	NEA
Input	water (river flow), own plants, power bought from IPPs, imports from India, staff, capital, tariff revenue
Process	generation → transmission (132/220/400 kV) → distribution → metering and billing
Output	electricity to domestic, industrial, commercial consumers; export in the wet season; revenue



NEA organization structure (simplified to directorate level).

2. Identification of the problem

- In FY 2072/73 NEA showed every sign of a failing public enterprise:
 - **load-shedding** up to **18 hours a day** in the dry winter;
 - **system loss 25.78%**: a quarter of energy lost, much of it theft;
 - accumulated **loss Rs 34.61 bn**; industries on diesel generators;
 - low staff morale; public anger.
- **Problem statement:** why did NEA perform so poorly, and what should it do in planning, leadership, motivation and HRD to improve?

3. Hypothesis

- **Main:** poor performance is as much a **management** failure as a shortage of generation.
- H1: planning horizon was short → fire-fighting, no storage.
- H2: leadership was weak and unstable → no accountability.
- H3: staff lacked motivators → leakage tolerated.
- H4: HRD was neglected → skill gaps in new technology.

4. Case research design

Data	Source and method
Secondary	NEA annual reports FY 2072/73–2080/81: loss, demand, supply, finance; news reports
Interviews	20 officers: MD office, 2 directorates, 6 distribution centres
Staff survey	200 staff, 5-point scale: pay, recognition, training, leadership
Consumer survey	300 households and 30 industries: supply hours, complaints
Observation	2 substations and 1 control centre (dispatch)
Analysis	trend charts; before/after 2073; fishbone of causes

5. Collection and analysis of data

- **System loss**, FY 72/73 to 80/81 (%): **25.78** → 22.90 → 20.45 → 15.32 → 15.27 → 17.18 (COVID year) → 15.38 → 13.46 → **12.73**.
- The sharpest fall came in the first three years after the new MD (2073 Bh).
- Each point of loss is roughly a billion rupees a year: NEA says leakage control earns it about Rs 12 bn a year.



Cause-and-effect analysis of NEA's performance in FY 2072/73.

(a) Planning horizon

- Found: planning mostly **short term**, one dry season at a time.
- Storage and peaking plants delayed for years → winter deficit grew with demand.
- Fix by three horizons:
 - **short** (1 yr): manage peak, cut leakage, import in winter;
 - **medium** (2–5 yr): transmission lines and substations, smart meters;
 - **long** (10–20 yr): storage projects, cross-border lines, export.
- H1 **accepted**.

(b) Leadership

- A new MD took charge on 14 Sept 2016 (Bhadra 2073).
- Within about two months Kathmandu Valley was declared load-shedding free (Tihar, Oct 2016); nationwide by **May 2018**.
- Mostly the **same** generation capacity: the change came from **dispatch management**, ending **dedicated-feeder** privileges, leakage raids, peak-hour imports.
- Style: high concern for task **and** people → 9,9 team management on Blake–Mouton's grid (§3.3); visible, decisive, accountable.
- H2 **accepted**: leadership was the turning factor.

6. Generalization and interpretation

- The turnaround came **before** big new plants: management, not only megawatts, explains most of it → main hypothesis supported.
- Supply growth (new plants, imports) then made the gains last.
- **Generalization**: other Nepali public enterprises (Nepal Telecom, Nepal Airlines, Nepal Oil Corporation) can improve the same way: stable accountable leadership, performance targets, a long planning horizon, merit-based HRD.
- **Limit**: the gains depend on one leader unless built into systems; political interference is the main risk.

(c) Motivation

- Staff survey (illustrative): pay and job security rated fair; recognition and growth rated poor.
- Herzberg (§3.2): **hygiene** factors present, **motivators** absent → no dissatisfaction, but no drive either.
- What worked: loss **targets per distribution centre**, monthly ranking, public recognition, action against poor performers.
- Maslow: esteem needs met by recognition; Vroom: effort → visible result → reward raised expectancy.
- H3 **accepted**.

(d) Human resource development

- Found: little structured training; promotion by seniority; no succession plan.
- New needs: smart metering, SCADA, grid operation, cross-border trade, project management.
- Needed: training-needs analysis (§2.4), skill-based training, merit rating linked to promotion, succession planning for key posts.
- H4 **partly accepted**: training exists but is not tied to career growth.

7. Conclusions and recommendations

Area	Recommendation	Horizon
Planning	rolling 1/5/20-year plan; load forecast; storage first	all three
Leadership	fixed MD tenure; performance contract with the Board	short
Motivation	loss and service targets per centre; ranking, rewards	short
HRD	needs analysis; training academy; merit promotion; succession	medium
Technology	smart meters, SCADA, MIS dashboards (§5)	medium
Governance	shield operations from political transfers	long

Checklist (step 6)

- **Achieves:** reliable supply, loss near 10%, sustained profit, motivated staff.
- **Difficulties:** union resistance to performance ranking; political transfers.
- **Limits:** capital for storage plants; dry-season hydrology; import capacity.
- **Best alternative?** Yes: cheaper and faster than building capacity alone.

Drawbacks of this study; references

- Survey figures are illustrative; a real study would sample properly.
- References: NEA annual reports; Kathmandu Post (2016, 2017, 2019); Khabarhub (2024).

4.5 Mock Case Study 2: Recommendations to Your College (74 Ch)

PIN 1/29 74 Ch · written out in full as a model · "XYZ Engineering Campus" and all its numbers are illustrative; put your own campus's facts in **verified/added**

Question (74 Ch, 8 marks)

Having spent more than 3 years in a particular college/campus of yours, what are the recommendations you wish to propose to the college management for the future improvement that would in turn boost up overall academic performance and image of the college?

1. Organizational background

- **XYZ Engineering Campus**, affiliated to TU, IOE; founded 2060 BS; private (company-owned) college.
- **Programmes:** BCT, BEI, BEL, BCE; about 960 students, 72 faculty (40 full time).
- **Goal:** produce competent engineers; top-10 IOE results; placement.
- **Structure:** Board → Principal → Heads of Department, Exam Section, Admin and Accounts (line and staff).
- **Financial position:** fee-funded; stable, little surplus for labs.

2. Identification of the problem

- Board pass rate fell from 74% to 62% over three years.
- Admissions fell 15%; few campus placements.
- **Problem:** what should management change to raise academic performance and the college's image?

3. Hypothesis

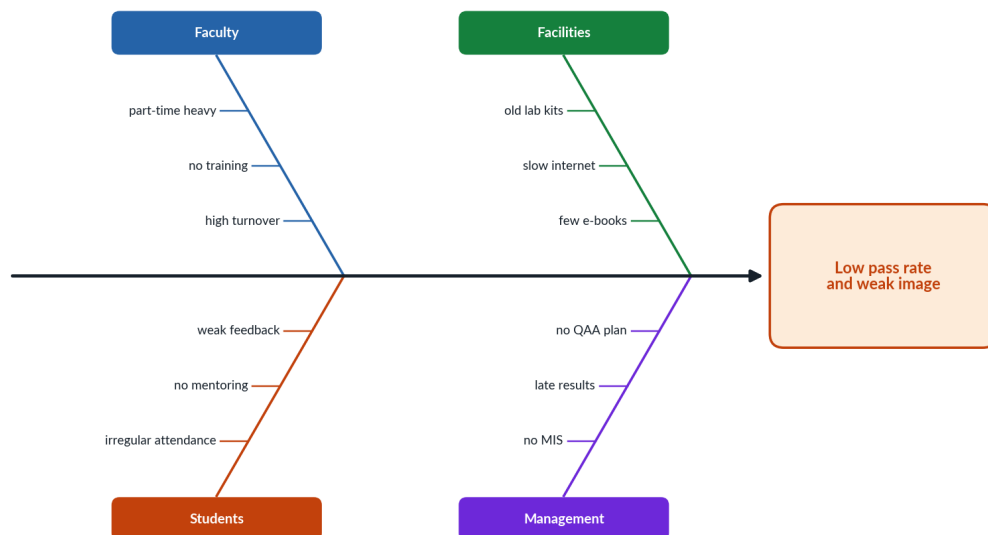
- Weak results come mainly from **faculty instability** and **poor facilities**, not from weaker students.

4. Case research design

Data	Source and method
Records	3 years of results, attendance, faculty turnover, admissions
Survey	240 students, 5-point scale on six areas
Interviews	12 faculty, 3 HoDs, the principal
Observation	labs, library, internet, hostel
Benchmark	two better-ranked IOE colleges

5. Collection and analysis of data

- Student survey, mean score out of 5:
 - Lowest: **career cell 1.8, internet 2.1, labs 2.3.**
 - Middling: library 2.9, hostel 2.6.
 - Strongest: teaching 3.4.
- Faculty turnover 25% a year; 45% of classes taken by part-timers.
- Students with over 80% attendance passed at 81%; the rest at 44%.



Cause-and-effect analysis: why results and image are falling.

6. Generalization and interpretation

- Attendance and faculty stability explain most of the pass-rate gap → hypothesis supported.
- Facilities (labs, internet) hurt practical courses and the college's image.
- Image follows results and placement: fix those, then publicize.
- Holds for most private affiliated colleges: **people and systems** before buildings.

Checklist

- Achieves:** higher pass rate, stronger admissions, better placement.
- Difficulties:** cost of full-time faculty; staff resistance to evaluation.
- Limits:** fee ceiling; TU calendar; space.
- Best alternative:** yes, most items cost little and repay through admissions.

7. Recommendations (by priority)

Area	Recommendation	When
Faculty	more full-time posts; training; merit-based pay and promotion	1 yr
Academic	attendance rule; mentoring; internal tests; remedial classes	now
Labs	replace kits; lab manuals; industry-sponsored labs	1–2 yr
Library, IT	e-journals; campus Wi-Fi; LMS for notes and assignments	1 yr
Students	career cell, internships, alumni talks, clubs	now
Research	small grants; MoUs with industry; project exhibitions	2 yr
Management	MIS for attendance, results, feedback; UGC QAA accreditation	1–3 yr
Image	website, social media, results and placement publicity	ongoing

Close: "Priorities 1–2 first; review every semester with student feedback."

4.6 Sample Case Study Report: Industrial Visit

internal assessment · **case study report = 10 of the 20 internal marks** · the course requires one after visiting an industrial organization · written out in full as a model · "ABC Cement Industries Ltd." and every number in it are illustrative: put your own visit's facts in **verified/added**

How to use this model

- Structure = the **7-step methodology** and the **report titles** of §4.3, merged into one report.
- Chapters 2–7 = the syllabus applied to one plant:
 - Ch. 1: ownership, structure, purchasing, marketing;
 - Ch. 2: manpower, recruitment, training, job evaluation, wages;
 - Ch. 3: motivation, leadership;
 - Ch. 5: MIS.
- Chapter 8 = the **case** proper: one problem, hypotheses, data, test.
- Real report: typed, 20–30 pages, figures and tables numbered, 2–3 visit photos if the plant allows.
- Keep every number traceable: say where it came from (log, report, survey).

Front matter order

- Title page.
- Acknowledgement.
- Abstract.
- Table of contents, list of figures and tables.
- List of abbreviations.

Table of contents, and the §4.3 step each chapter fills

Chapter	Contents	Step
1 Introduction	background of visit, objectives, methodology, limitations	4
2 Organization profile	history, objectives, products, capacity, process	1
3 Ownership, structure	public limited company, org chart	1
4 Personnel management	manpower, recruitment, training, job evaluation, merit rating, wages	5
5 Motivation, leadership	employee survey, theories applied	5
6 Purchasing, marketing	materials, spares, channels, promotion	5
7 MIS	reports by level, software	5
8 The case problem	problem, hypotheses, stoppage analysis, test	2, 3, 5
9 Findings	strengths, drawbacks, generalization	6
10 Recommendations	action table, plan, checklist	7
11 Conclusion	summary; references; appendices	7

Steps: 1 background, 2 problem, 3 hypothesis, 4 research design, 5 data and analysis, 6 generalization, 7 conclusions and recommendations.

Title page

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
[Campus name]

A Case Study Report on ABC Cement Industries Ltd.

Makwanpur, Nepal

Submitted as a requirement of
Organization and Management (ME 708)

Submitted by	Submitted to
[Name], [Roll no.]	Department of [...]
[Name], [Roll no.]	[Campus]
[Date of submission]	

Acknowledgement

We thank the management of ABC Cement Industries Ltd. for permission to visit the plant on [date], and the Plant Manager, the HR Officer, the Stores Officer and the shift engineers who guided us and answered our questions. We are grateful to [teacher], Department of [...], for guidance throughout, and to the campus for arranging the visit.

Abstract

This report studies ABC Cement Industries Ltd., an integrated cement plant with a 1,200 TPD kiln. It describes the company's ownership, structure, personnel practices, motivation, leadership, marketing and MIS. The case problem is low capacity use: 64–71% over four years. Kiln logs for FY 2082/83 show 1,490 hours of unplanned stops, 69.1% of them from power trips and breakdowns. The report recommends a standby power supply, preventive maintenance, spares planning, training and regularizing contract operators, and an ERP, aiming at 80% capacity use in two years.

Chapter 1. Introduction

1.1 Background of the visit

- One-day visit on [date]: 32 students of IV-I and 2 teachers.
- Why this plant: a **continuous-process** industry → every O&M topic is visible: structure, shifts, manpower, maintenance, marketing.

1.2 Objectives of the study

1. Study the organization: history, ownership, structure.
2. Study personnel practices: manpower planning to wages.
3. Study motivation, leadership and MIS in practice.
4. Identify the plant's main management problem, analyse its causes, recommend solutions.

1.4 Limitations

- One-day visit; sample of 60 out of 412 employees.
- Some cost and wage data confidential → published figures only.
- Respondents chosen by the HR section → possible bias.

1.3 Methodology (case research design)

Method	What and from whom
Observation	plant tour: quarry, crusher, kiln, central control room, packing, stores
Interviews	Plant Manager, HR Officer, Stores Officer, Marketing Officer, 2 shift engineers
Questionnaire	60 employees (40 permanent, 20 contract), 5-point scale; Appendix A
Records	annual reports FY 2079/80–2082/83, kiln stop-page log, maintenance log, HR records
Analysis	tables, bar chart, cause-and-effect (fishbone) diagram, hypothesis test

Chapter 2. Organization Profile

2.1 History

- **2065 BS:** registered as a private limited company (Companies Act 2063).
- **2068:** commercial production, grinding purchased clinker.
- **2072:** own 1,200 TPD kiln commissioned → integrated plant.
- **2076:** converted to public limited; IPO of 30% of shares; listed on NEPSE.
- **2080:** cement mill expanded to 1,500 TPD.

2.2 Vision and objectives

- **Vision:** most trusted cement brand of central Nepal.
- **Objectives:**
 - produce cement to Nepal Standard (NS) quality;
 - run the plant at 85% of capacity or more;
 - earn profit, pay regular dividend;
 - employ local people, substitute imports.

2.4 Input, process, output

Stage	ABC Cement
Input	limestone (own quarry, 12 km), clay, iron ore, gypsum and coal (imported via Birgunj), fly ash, power (NEA grid, 33 kV), water, 412 staff, capital
Process	quarry → crusher → raw mill → preheater and kiln (1,450 °C) → clinker → cement mill → packing (figure)
Output	OPC and PPC in 50 kg bags; dust (held by bag filters), waste heat; revenue, taxes, jobs

2.3 Products, capacity, performance

Item	Value
Products	OPC (53 grade), PPC; brand “ABC Strong”
Clinker capacity	1,200 TPD = 396,000 t a year (330 days)
Cement capacity	1,500 TPD = 495,000 t a year
Employees	412 (248 permanent, 164 contract)
Working	3 shifts of 8 h; kiln runs round the clock
Finance, FY 82/83	sales Rs 4.2 bn; net profit Rs 180 million; 10% cash dividend

FY (BS)	Cement, '000 t	Capacity used
2079/80	351	70.9%
2080/81	332	67.1%
2081/82	318	64.2%
2082/83	342	69.1%

Capacity used = production ÷ 495,000 t.

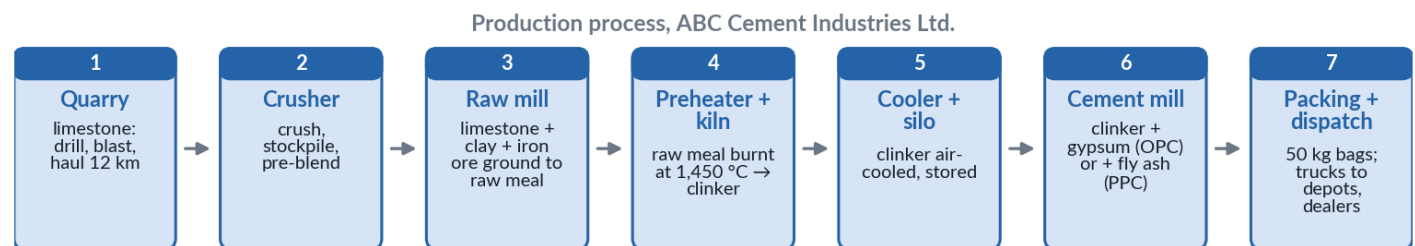


Figure 2.1: Dry-process production flow of the plant.

Chapter 3. Form of Ownership and Organization Structure

3.1 Form of ownership: public limited company (§1.4)

- Registered with the OCR under the Companies Act 2063; listed on NEPSE.
- Shares: promoters 70%, general public 30%.
- Separate legal entity, limited liability, perpetual succession.
- AGM elects a Board of 7 directors (5 promoter, 2 public).
- Board appoints the MD as chief executive.
- Disclosure to SEBON and NEPSE; yearly external audit.
- Suits the plant: large capital raised from the public, professional management.

3.2 Organization structure (§1.5)

- **Type:** line and staff.
 - line: MD → Plant Manager → department heads → shift engineers → supervisors → operators;
 - staff: Safety and Environment advises the MD (dashed in the chart).
- **Departmentation:** by function.
- **Span of control:** MD 7; shift engineer about 12 operators.
- **Centralization:** purchases above Rs 5 lakh and all hiring decided by the MD; shift decisions delegated.
- **Levels:** 6, from Board to operator.

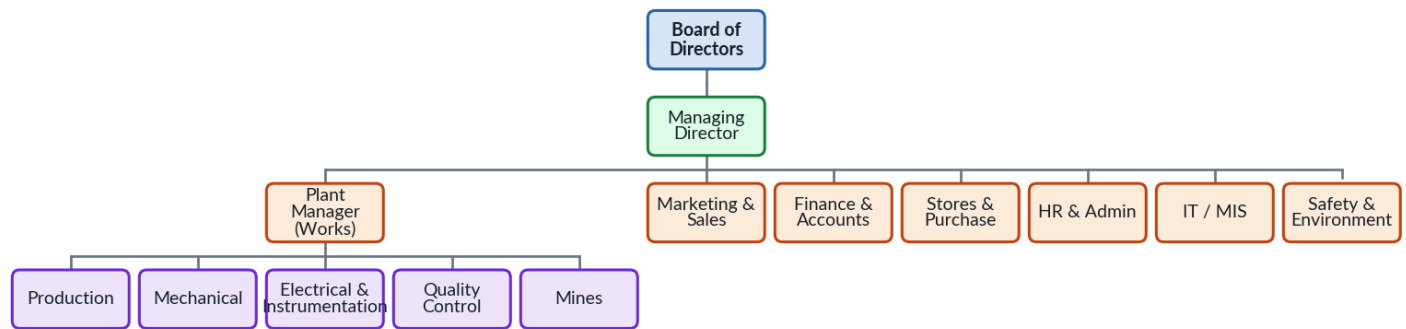


Figure 3.1: Organization chart of ABC Cement Industries Ltd.

Chapter 4. Personnel Management

4.1 Personnel policy and manpower planning (§2.2, §2.3)

- Written HR bylaws under the Labour Act 2074:
 - 8 h a day, 48 h a week;
 - overtime at 1.5 times the wage rate;
 - probation up to 6 months.
- Manpower plan made yearly with the budget, from the production target.
- No long-term forecast: posts filled only after a vacancy arises.

4.2 Recruitment and selection (§2.4)

- **Internal:** promotion to supervisor and officer posts.
- **External:**
 - newspaper and website advertisements;
 - campus (IOE) for trainee engineers;
 - employee referrals;
 - labour contractor for packing and loading labour (contract).
- **Selection:** screening → written test (engineers, officers) → panel interview (Plant Manager, HR, department head) → medical test → appointment → 6-month probation.
- 60% of staff from Makwanpur, per agreement with local community.

Manpower, FY 2082/83

Department	Staff
Production	142
Mechanical, Electrical and Instrumentation	88
Mines (quarry)	46
HR and Admin (incl. security, drivers)	32
Marketing and Sales	26
Quality Control (laboratory)	24
Finance and Accounts	20
Stores and Purchase	18
Safety and Environment	8
IT / MIS	8
Total	412
Level	Staff
Top (MD, Plant Manager, GMs)	6
Middle (department heads, engineers, officers)	38
Supervisory	72
Operators and workers	296

4.3 Training and development (§2.5)

- **Induction:** 3 days, including safety.
- **On the job:** new operator works under a senior operator for 3 months.
- **Off the job:** equipment supplier trains engineers on the kiln and DCS; plant visits abroad for 2 engineers a year.
- FY 2082/83: 14 programmes, 186 participants.
- Contract workers: safety briefing only.
- No training-needs analysis, no skill matrix.

4.4 Job evaluation and merit rating (§2.6)

- **Job evaluation:** grading method → 10 grades, G1 (helper) to G10 (department head).
- Written job descriptions for officer posts only.
- **Merit rating:** yearly, graphic rating scale, rated by immediate supervisor on 6 factors:
 - quality and quantity of work;
 - attendance, safety;
 - teamwork, initiative.
- Used for the annual increment only: promotion mostly by seniority.

4.5 Wages, incentives and welfare (§2.7)

- **Time wage:** monthly salary by grade + yearly grade increment.
- **Allowances:** night-shift, dearness, festival allowance of one month's basic salary (Dashain).
- **Production incentive:** paid when monthly output exceeds 90% of target → earned in only 3 of 12 months of FY 2082/83.
- **Bonus:** 10% of net profit under the Bonus Act 2030 → Rs 18 million shared in FY 2082/83.
- **Social security:** SSF contributions under the Contribution-based Social Security Act 2074; medical insurance.
- **Welfare:** canteen, staff bus, quarters for 90 families.
- Contract workers paid by the contractor at about the minimum wage, few benefits.

4.6 Industrial relations (§2.8)

- 2 unions of permanent staff; collective agreement every 2 years.
- 3-day strike in 2081 over regularizing contract workers → settled with a promise to review.

Chapter 5. Motivation and Leadership

5.1 Employee survey (Appendix A)

Factor (mean, 1-5)	Permanent (40)	Contract (20)
Job security	4.1	1.9
Pay and allowances	3.4	2.6
Supervision	3.3	2.9
Working conditions, safety	3.2	2.8
Training and growth	2.7	1.6
Recognition	2.4	2.0
HR records	Permanent	Contract
Turnover a year	4%	28%
Absenteeism	3%	9%

5.2 Analysis (§3.2)

- **Herzberg:**
 - permanent staff: hygiene factors met (security 4.1, pay 3.4), motivators weak (recognition 2.4, growth 2.7) → not dissatisfied, but not driven;
 - contract staff: even hygiene factors unmet (security 1.9) → dissatisfaction → 28% turnover.
- **Maslow:** contract workers stuck at safety needs; permanent staff seek esteem.
- **Vroom:** incentive earned in 3 months of 12 → effort not seen to bring reward → low expectancy.

5.3 Leadership (§3.3)

- **MD:** sets targets, decides fast; autocratic on purchase and hiring, participative in the monthly production review.
- **Shift engineers:** task-oriented, 9,1 on Blake-Mouton's grid: output first, little coaching.
- **Plant Manager:** visits every shift, listens; rated highest by staff → closest to 9,9 team management.

Chapter 6. Purchasing and Marketing (§1.6)

6.1 Purchasing and stores

- **Limestone:** own quarry, reserve for about 30 years.
- **Clay, iron ore:** local suppliers on yearly contract.
- **Coal:** imported by tender from 2 suppliers, lead time 6–8 weeks.
 - stock target 45 days;
 - fell to 12 days twice in FY 2082/83 → kiln slowed.
- **Spares:** mostly imported, lead time 3–4 months.
 - bought only after a breakdown: no minimum stock → long stops.
- Purchase committee for orders above Rs 5 lakh; quotations or tender.

6.2 Marketing

- **Channel:** plant → 4 depots → 180 dealers → retailers → users.
- **Direct sales** to projects (hydropower, roads, bridges): 20% of volume.
- **Market:** Nepal self-sufficient in cement, many plants → oversupply, price competition.
- **Price:** follows market leader; volume discounts to dealers.
- **Promotion:** hoardings, FM radio, dealer meets, mason training programmes.
- **Seasonality:** demand falls in monsoon (Ashad to Bhadra) and at Dashain–Tihar → clinker yard fills → kiln stopped 240 h.

Chapter 7. Management Information System (Ch. 5)

7.1 Reports by level

Level	Report	How often
Operational	shift log, DCS trends, stores issue slips	every shift
Middle	daily production report (DPR), weekly maintenance report	daily, weekly
Top	monthly MIS report to MD; board report; annual report	monthly, quarterly, yearly

7.2 Findings

- DCS in the central control room logs every kiln parameter.
- Accounts on accounting software; stores, maintenance and sales on separate spreadsheets.
- No link → maintenance cannot see spares stock → learns of a shortage only after a breakdown.
- DPR reaches the MD at 10 am next day: no real-time KPI.

Chapter 8. The Case Problem: Low Capacity Use

8.1 Identification of the problem

- Capacity used 64–71% for four years (section 2.3), against the 85% the budget assumes.
- Every idle kiln hour loses 50 t of clinker (1,200 TPD ÷ 24): the loss cannot be made up later.
- **Problem statement:** why does the plant use only about two-thirds of its capacity, and what should management do?

8.2 Hypotheses

- **H1:** unplanned kiln stops from power trips and breakdowns are the main cause.
- **H2:** weak off-season demand adds to it.
- **H3:** manpower practices (contract turnover, little training) cause many of the breakdowns.
- **H4:** raw-material shortage is the main cause.

8.3 Data: kiln stops, FY 2082/83

Cause	Hours	% of unplanned
Power trips, voltage dips	610	40.9
Mechanical breakdowns	420	28.2
Clinker yard full (market)	240	16.1
Coal, material shortage	150	10.1
Others	70	4.7
Unplanned total	1,490	100.0
Planned shutdown	360	
All stops	1,850	

8.4 Analysis

- Kiln ran $8,760 - 1,850 = 6,910$ h $\rightarrow$ run factor 78.9%.
- Clinker lost to unplanned stops: $1,490$ h $\times 50$ t = $74,500$ t = 18.8% of the 396,000 t capacity.
- Power + breakdowns = $1,030$ h = **69.1%** of unplanned stops.
- **Power:** 146 trips in the year.
 - each trip $\rightarrow$ about 4 h to restart and stabilize the kiln;
 - a 5-minute trip costs hours of output.
- **Breakdowns:** 29 in the maintenance log.
 - 11 traced to operating error (missed lubrication, overloading);
 - 9 of those 11 on shifts run mainly by contract operators;
 - 6 waited for imported spares $\rightarrow$ 180 of the 420 h.
- **Market:** yard-full stops all fell in monsoon and festival months.

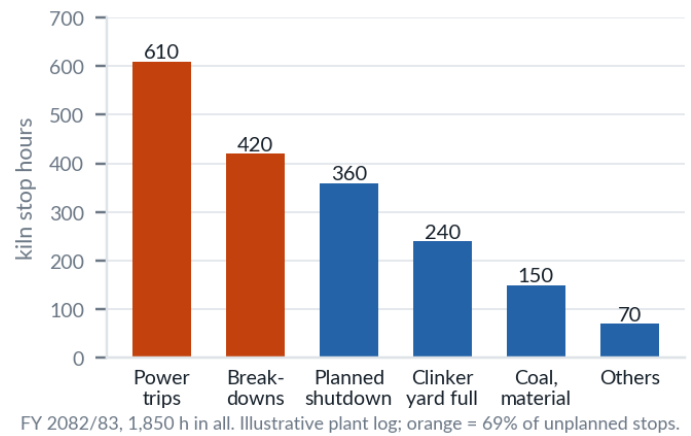


Figure 8.1: Kiln stop hours by cause.

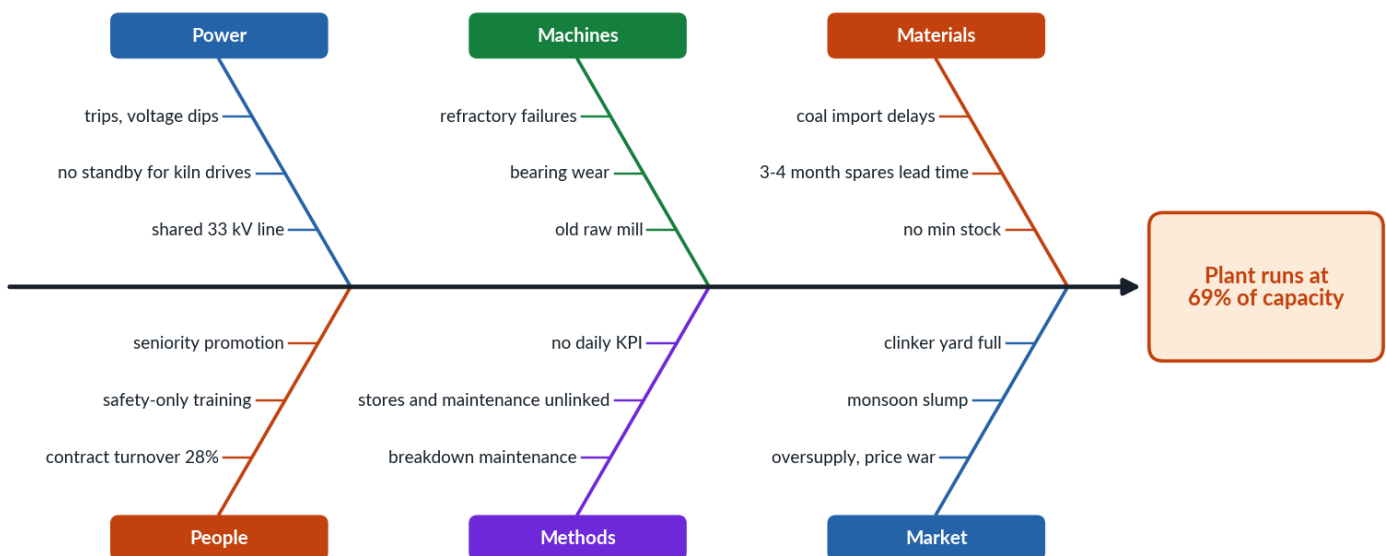


Figure 8.2: Cause-and-effect analysis of low capacity use.

8.5 Testing the hypotheses

H	Evidence	Result
H1	power trips + breakdowns = 1,030 h = 69.1% of unplanned stops	accepted
H2	clinker yard full 240 h = 16.1%, all in monsoon and festival months	partly accepted
H3	11 of 29 breakdowns from operating error, 9 of them on contract-run shifts; contract turnover 28%	accepted
H4	material shortage 150 h = 10.1%	rejected as main cause

Chapter 9. Findings, Generalization and Interpretation

9.1 Strengths

- Integrated plant with own quarry → low raw-material cost.
- NS-marked product; brand known in 30 districts.
- Profitable; regular 10% dividend.
- Good pay and welfare for permanent staff; lawful HR bylaws.
- Modern kiln with DCS control.

9.3 Generalization and interpretation

- Capacity is lost mainly **inside the plant gate** (69.1% of unplanned stops), not in the market → a **management** problem.
- Holds for continuous-process plants (cement, steel, paper): a lost kiln hour is never recovered → reliability of power, maintenance and people decides output.
- Limit: one plant, one year of logs.

9.2 Drawbacks (problems found)

1. Breakdown maintenance instead of preventive.
2. No standby power for kiln drives.
3. Spares bought after failure, not stocked.
4. Two-tier workforce: contract operators insecure, untrained, 28% turnover.
5. Promotion by seniority; merit rating used for increments only.
6. Production incentive too hard to earn (3 of 12 months).
7. MIS not integrated; no real-time KPI.
8. No off-season marketing plan.

Chapter 10. Recommendations

Area	Recommendation	Horizon	Responsible
Power	dedicated industrial line from NEA; standby DG set for kiln drive and fans; voltage stabilization	short to medium	Electrical
Maintenance	preventive schedule; vibration monitoring and kiln shell scanner; annual shutdown in monsoon, when demand is low anyway	short	Mechanical
Stores	VED and ABC analysis of spares; min-max stock of critical spares; rate contracts with suppliers	short	Stores and Purchase
People	training-needs analysis, skill matrix, operator certification; regularize long-serving contract operators; merit-weighted promotion	medium	HR
Motivation	shift incentive on kiln run hours; best-shift recognition; suggestion scheme	short	HR, Plant Manager
MIS	ERP linking stores, maintenance, production and sales; daily KPI dashboard	medium	IT / MIS
Marketing	new districts, project sales, pre-monsoon dealer stocking schemes, mason meets	medium	Marketing

Expected result

- Unplanned stops 1,490 h → under 800 h a year.
- Capacity used 69.1% → **80%** within two years.

Checklist (step 6)

- **Achieves:** about 80% capacity use, fewer breakdowns, stable workforce.
- **Difficulties:** capital for DG set and ERP; contractor and union resistance.
- **Limits:** NEA grid capacity, market demand, import lead times.
- **Best alternative?** Yes: more output from the existing plant costs far less than new capacity.

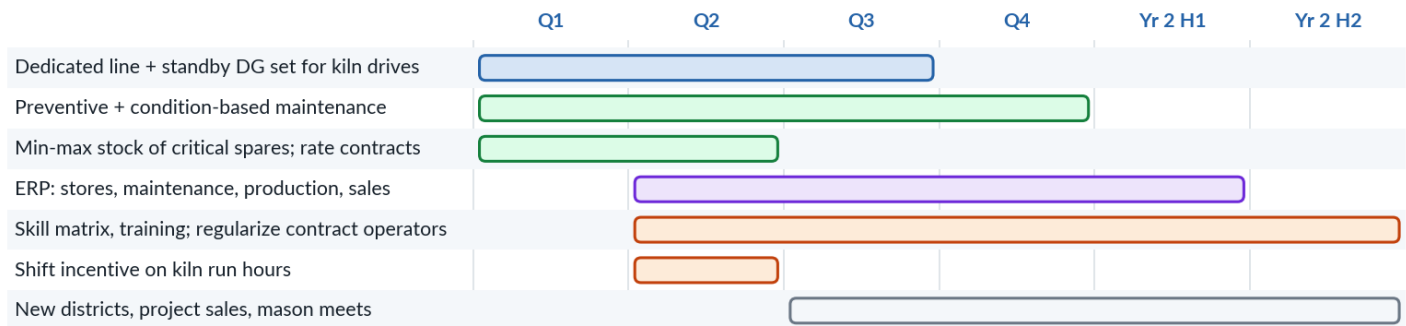


Figure 10.1: Implementation plan over two years.

Chapter 11. Conclusion

- ABC Cement = profitable public limited company, line-and-staff structure, functional departments.
- Personnel practices meet the law for permanent staff; the contract workforce is left out.
- Main problem: only about 69% of capacity used.
 - caused mainly by unplanned kiln stops (power, breakdowns);
 - not by demand or materials.
- Remedy lies in management: preventive maintenance, spares planning, trained and stable operators, integrated MIS.
- The visit showed O&M theory at work in a real plant.

References

1. ABC Cement Industries Ltd., Annual Reports FY 2079/80–2082/83.
2. Government of Nepal: Labour Act 2074; Bonus Act 2030; Companies Act 2063; Contribution-based Social Security Act 2074.
3. Adhikari, B., *Organization and Management* slide notes.
4. Lecture notes, Organization and Management (ME 708), IOE.

Appendix A: Questionnaire (sample)

- **Part A:** department; permanent or contract; years of service.
- **Part B:** rate 1 (very poor) to 5 (very good):
 - job security; pay and allowances;
 - supervision; working conditions and safety;
 - training and growth; recognition.
- **Part C (open):**
 - What stops the kiln most often, in your experience?
 - What one change would most improve your work?

Appendix B: Interview checklist

- **Plant Manager:** capacity, stops, maintenance practice.
- **HR Officer:** manpower, recruitment, training, appraisal, wages.
- **Stores Officer:** lead times, stock policy.
- **Marketing Officer:** channels, prices, seasons.
- **IT Officer:** reports, software.

4.7 PYQ Mapping

Theory questions, covered in this document

- **What is a case study, why conducted, objectives, value** → §4.1. 8 papers; three of them as a 4-mark short note.
- **Types of case studies** 80 Ba 2 → §4.1.
- **Phases and steps, with an example from your field** → §4.2. 82 Bh 4, 81 Bh 4, 81 Ba 5, 80 Bh 8, 79 Bh 8
- **Prepare a case study of NEA / Nepal Telecom** 72 Ch 8 → §4.3 structure, §4.4 full model.
- **Recommendations for your college** 74 Ch 8 → §4.3 structure, §4.5 full model.
- **Case study report** (internal assessment, 10 marks, not examined) → §4.6 complete sample.
- 73 Ch and 71 Ch pair case-study objectives with MIS needs; the MIS half is in Ch. 5.

Numerical content

- This chapter has no numerical component.

Sources

- Adhikari pp259–272 and AJ Sir's 7-slide deck cover all five syllabus topics between them. S.K. Joshi's textbook follows an older syllabus and has no case-study chapter, so nothing was taken from it here.
- No local source has a case-study diagram: every figure in this chapter is generated (ch4fig.py). Only the process flows, the NEA org chart and the two fishbones are kept: the diagrams an answer would draw.

- NEA facts: NEA annual reports as reported by Kathmandu Post (16 Aug 2016, 18 Aug 2017, 15 Feb 2019) and Khabarhub (Nov 2024).

Abbreviations, chips and month codes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
JSC	joint stock company	MNC	multinational company
HR, HRM	human resource (management)	OB	organizational behaviour
MBO	management by objectives	CEO, MD	chief executive officer, managing director
AGM	annual general meeting	ESOP	employee share ownership plan
MoA, AoA	memorandum / articles of association	OCR	Office of the Company Registrar
SEBON	Securities Board of Nepal	NEPSE	Nepal Stock Exchange
IPO	initial public offering	PAN, VAT	permanent account no., value added tax
NEA	Nepal Electricity Authority	NOC	Nepal Oil Corporation
QC	quality control	R&D	research and development
OR	operations research	ROI	return on investment
CIM	computer integrated manufacturing	FMS	flexible manufacturing system
CAD, CAM	computer aided design / manufacturing	CAPP	computer aided process planning
CNC	computer numerical control	AGV	automated guided vehicle
MIS	management information system	IT	information technology
SaaS	software as a service	BPO	business process outsourcing
CSR	corporate social responsibility	POSDCoRB	plan, organize, staff, direct, coordinate, report, budget
OPC, PPC	ordinary Portland / Portland pozzolana cement	TPD	tonnes per day
DCS	distributed control system	ERP	enterprise resource planning
DG	diesel generator	KPI	key performance indicator
DPR	daily production report	SSF	Social Security Fund
NS	Nepal Standard (quality mark)	VED	vital, essential, desirable (spares)

Short	Means	Short	Means
no.	number	info	information
vs, Vs	versus	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and
=	is / equals	min, max	minimum, maximum
Ch.	chapter	§	section

Chip	Means (29 papers, 2065–2082 BS)
TOP n/29	asked in 8 or more papers
HOT n/29	asked in 4 to 7 papers
PIN n/29	asked in 1 to 3 papers, inside an 8-mark question
LOW n/29	asked in 1 to 3 papers, as a 4-mark short note only
syllabus, no PYQ yet	in the syllabus, never examined
4+4	marks as printed; several chips = the question recurred at different marks
verified/added	researched and added beyond the lecture notes
80 Bh	bold year = Regular exam; plain = Back exam

Months: Ba = Baishakh, Jth = Jestha, Asa = Ashad, Shr = Shrawan, Bh = Bhadra, Ash = Ashwin, Ka = Kartik, Mng = Mangsir, Po = Poush, Ma = Magh, Ch = Chaitra. **Ash and Asa are different months.** **Sources:** AJ Sir = AJ Sir's lecture decks; Adhikari = Bikash Adhikari, *Organization and Management* slide notes; Joshi = S.K. Joshi, *Organization and Management*, A.K. Books.